

Eric Waters

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Software Engineer with an M.S. in Big Data Systems and 3+ years of experience at Intel and Dell. Skilled in both backend and frontend, with extensive experience building APIs, responsive UIs, data processing pipelines, AI/ML integrations, and cloud/DevOps solutions.

Experience

Intel Corporation

Chandler, Arizona

Software Application Development Engineer

August 2023 – Present

Offline Data Analytics Platform

- Architected and launched a centralized Offline Analytics Platform, integrating Airflow, Kubernetes, CI/CD, and ELK to standardize ETL development and eliminate duplicated tooling across teams.
- Reduced ETL onboarding time from 2 weeks to 2 days by delivering reusable pipelines, preconfigured infrastructure, and Python libraries for rapid DAG development and logging.
- Drove cross-team adoption and alignment by partnering with platform teams and establishing shared standards, improving scalability, observability, and readiness for enterprise data migration.

Generative AI Web Platform

- Developed a suite of 5 GenAI applications using Angular and Python (Sanic), including a legal contract audit tool estimated to save \$30M annually (recipient of a Divisional Recognition Award).
- Architected and designed a no-code drag-and-drop AI app studio using React, Next.js, Tailwind CSS, enabling non-technical teams to build custom AI workflows in record time.
- Redesigned the platform's UI/UX, implementing reusable frontend components to standardize design language and accelerate development cycles across 6+ developers.

Factory Excursion Management API

- Spearheaded the architectural merger of two massive FastAPI instances, unifying authentication (OAuth 2.0/OpenID), logging, and CI/CD pipelines to reduce long-term maintenance overhead.

Factory ML Analysis API

- Engineered a high-availability FastAPI service to expose legacy .NET ML algorithms to external applications, handling hundreds of thousands of requests.
- Implemented a Blue-Green deployment strategy on Windows VMs using reverse proxies and GitHub Actions, ensuring zero-downtime updates for factory analysis.

AI Cognitive Search Engine

- Sole Administrator for a cross-company cognitive search engine; developed ETL pipelines using Python, Pandas, and SQL Server to index millions of documents from SharePoint, Wikis, and other internal manufacturing document sources engine.

Dell Technologies

Round Rock, Texas

Software Engineering Intern (Undergraduate & Graduate)

May 2021 – August 2021 & June 2022 – August 2022

- Implemented OAuth 2.0 SSO and fine-grained authorization across enterprise apps and APIs using Spring Boot and Angular, enhancing security, compliance, and user experience.
- Built a custom Kanban board app with Spring and MySQL, improving agile collaboration for engineering teams.

Education

Arizona State University

Tempe, Arizona

Computer Science (Big Data Systems), M.S. with Distinction

4.0 GPA | May 2023

Computer Science, B.S. Summa Cum Laude & Barrett, the Honors College

4.0 GPA | May 2022

Skills

- **Programming Languages:** Python, Java, Typescript, JavaScript, C/C++, SQL
- **Frontend:** React, Next.js, Angular, Tailwind CSS
- **Backend:** FastAPI, Sanic, Spring Boot, REST, OAuth 2.0
- **Data & ETL:** Pandas, Airflow, Kafka, Spark, PostgreSQL, SQL Server, MySQL
- **DevOps & Cloud:** Docker, Kubernetes, Rancher, GitHub Actions, ELK Stack

Projects

Distributed Graph Processing Pipeline for Real-Time Taxi Data

May 2023

- Architected a distributed data pipeline using Kubernetes and Kafka to ingest and process taxi geospatial datasets in real-time.
- Engineered a scalable graph database solution using Neo4j and Kafka Connect, implementing PageRank and Breadth-First Search algorithms to identify critical transit hubs and pickup locations.